

Fig. 2: Actual-size PCB layout of 5V DC converter with LM2574

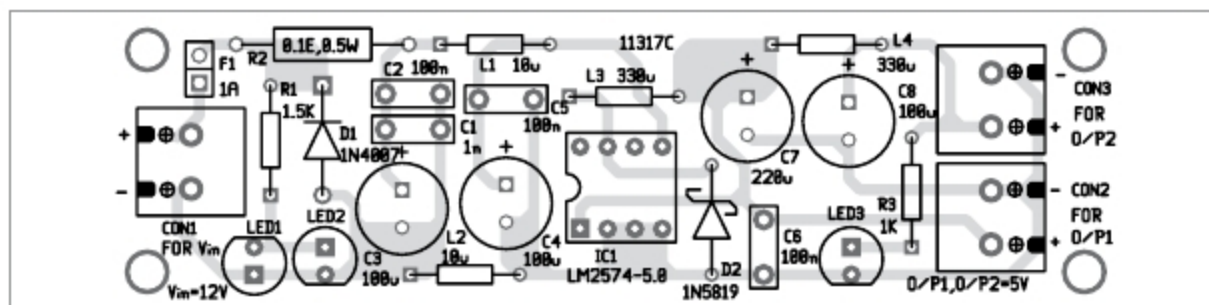


Fig. 3: Components layout for the PCB

This converter may generate noise. Hence, an input LC filter is used to suppress the noise. Capacitors C1, C2, C3 and inductors L1 and L2 are part of the input composite LC filter. L1 and L2 are wound on the same core, and are used to prevent input ripples and noise. It is possible to use two separate inductors, but that will give lower

filtering capabilities. These inductors are selected according to maximum output current.

C4 and C5 provide additional filtering and reservoir capabilities for the switching IC.

Output voltage is 5V and output current is up to 0.5A. Switching frequency of LM2574 is around 52kHz,

which is higher than the audible frequency range.

L3, D2, C6 and C7 convert pulses from LM2574 to 5V DC voltage with some ripples, which are available at output connector CON2. Further filtering can be done using L4 and C8. This output is made available at CON3.

LED3 shows the presence of 5V DC output across CON2 and CON3.

Construction and testing

An actual-size PCB layout of the low-noise 5V DC converter with LM2574 is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on the PCB, connect 12V input across CON1. 5V output is available across CON2 and CON3. The circuit does not need any adjustments. **EFY**



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